

Diodes

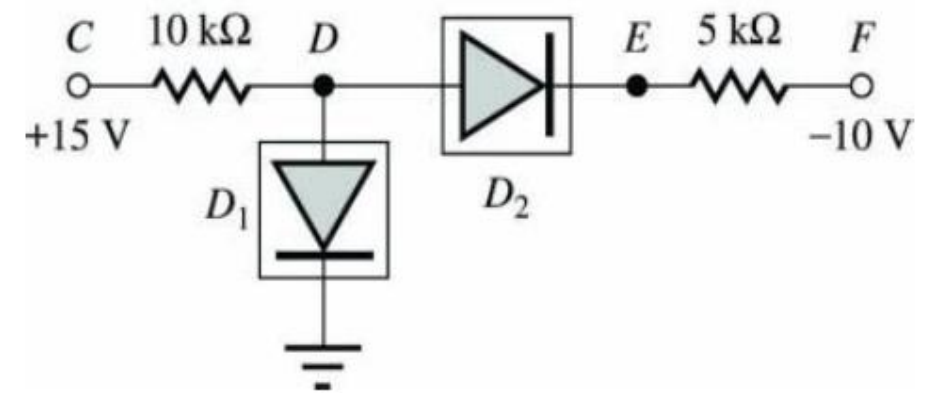
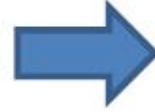
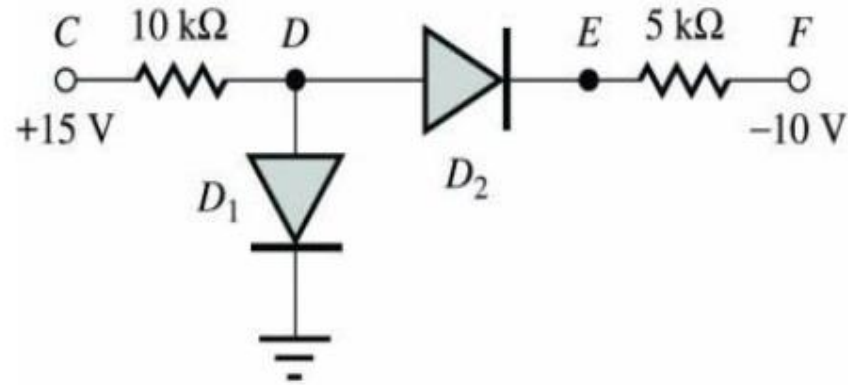
EXAMPLE

Two-Diode Circuit Analysis

5 steps to solve diode circuits

1. Select a diode model
2. Label i_D and v_D
3. Guess each diode's region of operation from circuit.
4. Analyze circuit using the equivalent circuit solve the currents and voltages at any node.
5. Verify that the equivalent replacements were valid. If not, guess another state for the diode.

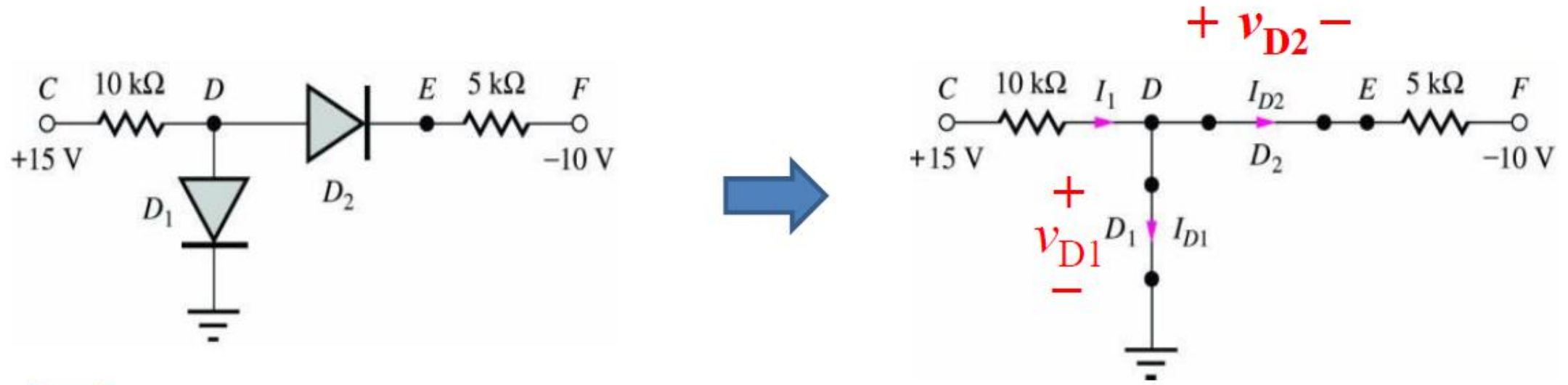
Two-Diode Circuit Analysis



1

The ideal diode model is chosen.

Two-Diode Circuit Analysis



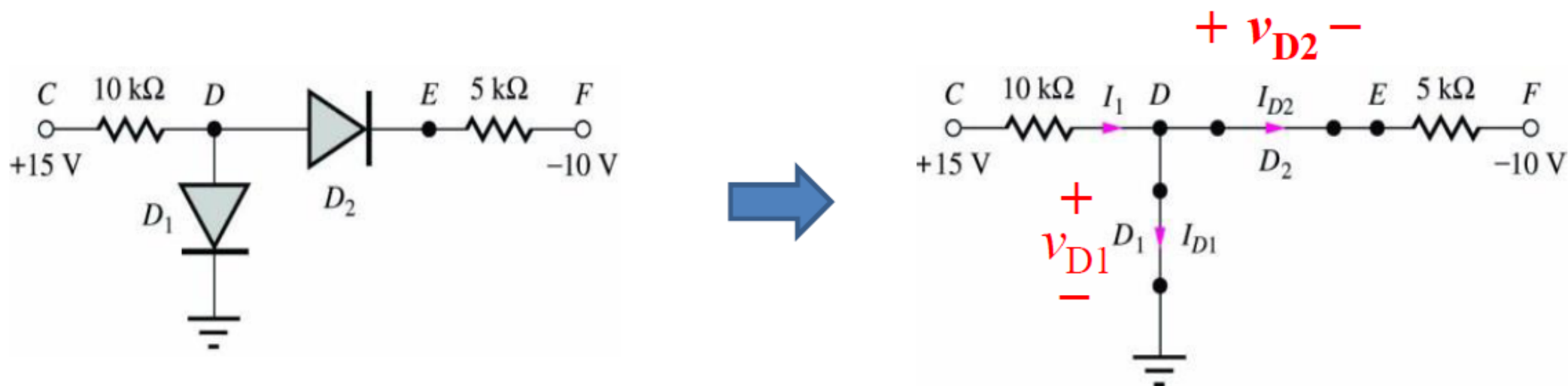
1

The ideal diode model is chosen.

2

Label i_D and v_D .

Two-Diode Circuit Analysis



1

The ideal diode model is chosen.

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Label i_D and v_D .

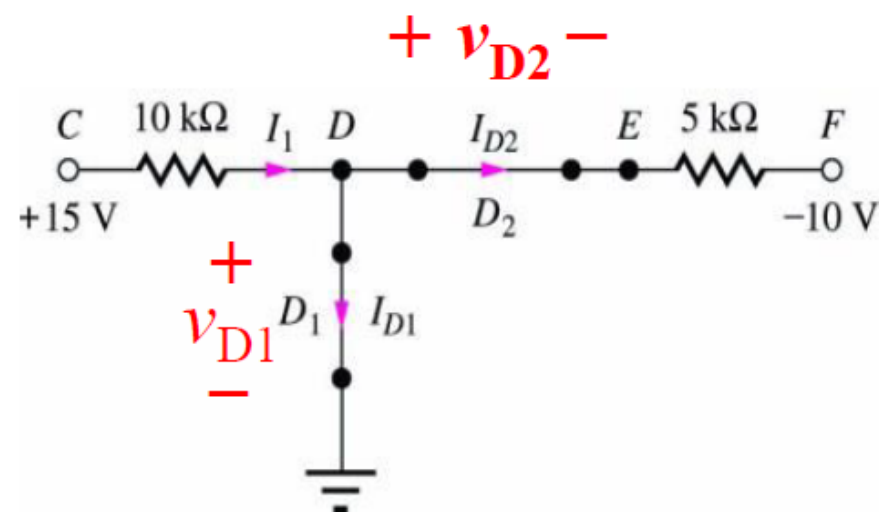
3

Guess an operation state of each diode.

Since the 15-V source appears to force positive current through D_1 and D_2 , and the -10-V source is forcing positive current through D_2 , **assume both diodes are on.**

→ i_{D1} and $i_{D2} > 0$.

Two-Diode Circuit Analysis



Analysis: Since the voltage at node D is zero due to the short circuit of ideal diode D_1 ,

$$I_1 = \frac{(15 - 0)V}{10k\Omega} = 1.50 \text{ mA}$$

$$I_{D2} = \frac{0 - (-10V)}{5k\Omega} = 2.00 \text{ mA}$$

$$I_1 = I_{D1} + I_{D2} \quad | \quad I_{D1} = 1.50 - 2.00 = -0.500 \text{ mA}$$

The Q-points are $(-0.5 \text{ mA}, 0 \text{ V})$ and $(2.0 \text{ mA}, 0 \text{ V})$

1

The ideal diode model is chosen.

2

Label i_D and v_D .

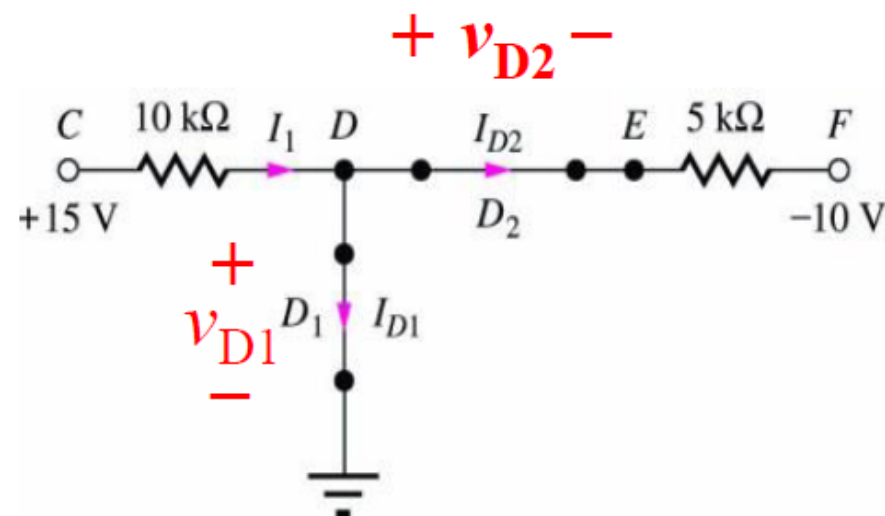
3

Guess an operation state.

4

Analyze circuits.

Two-Diode Circuit Analysis



1

The ideal diode model is chosen.

2

Label i_D and v_D .

3

Guess an operation state.

4

Analyze circuits.

5

Verify circuits.

Analysis: Since the voltage at node D is zero due to the short circuit of ideal diode D_1 ,

$$I_1 = \frac{(15 - 0)V}{10k\Omega} = 1.50 \text{ mA}$$

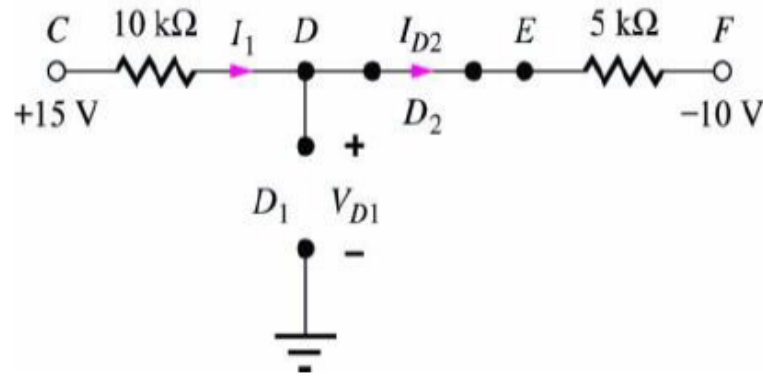
$$I_{D2} = \frac{0 - (-10V)}{5k\Omega} = 2.00 \text{ mA}$$

$$I_1 = I_{D1} + I_{D2} \quad | \quad I_{D1} = 1.50 - 2.00 = -0.500 \text{ mA}$$

The Q-points are $(-0.5 \text{ mA}, 0 \text{ V})$ and $(2.0 \text{ mA}, 0 \text{ V})$

But, $I_{D1} < 0$ is not allowed by the diode, so try again.

Two-Diode Circuit Analysis



Analysis: Since current in D_2 is valid, but that in D_1 is not, the second guess is D_1 off and D_2 on,

$$15 - 10,000I_1 - 5,000I_{D2} - (-10) = 0$$

$$I_1 = \frac{25V}{15,000\Omega} = 1.67 \text{ mA}$$

$$V_{D1} = 15 - 10,000I_1 = 15 - 16.7 = -1.67 \text{ V}$$

1

The ideal diode model is chosen.

2

Label i_D and v_D .

3

Guess an operation state.

4

Analyze circuits.

5

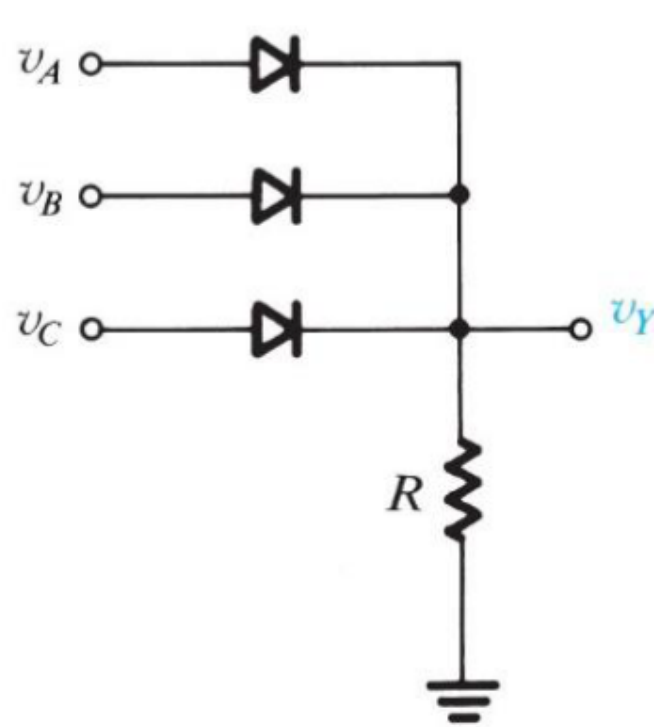
Verify circuits.

Q-Points are D_1 : (0 mA, -1.67 V): off

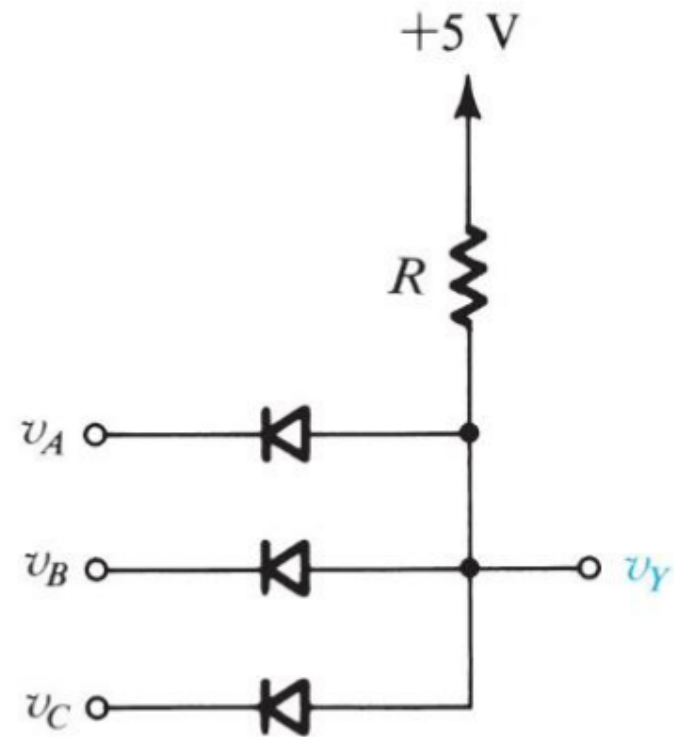
D_2 : (1.67 mA, 0 V) : on

Now, the results are consistent with the assumptions.

Diode logic gates



(a) $Y=A+B+C$



(b) $Y=A \cdot B \cdot C$

Figure 4.5 Diode logic gates: (a) OR gate; (b) AND gate (in a positive-logic system).

Limiting and Clamping Circuits

Rectifier circuit

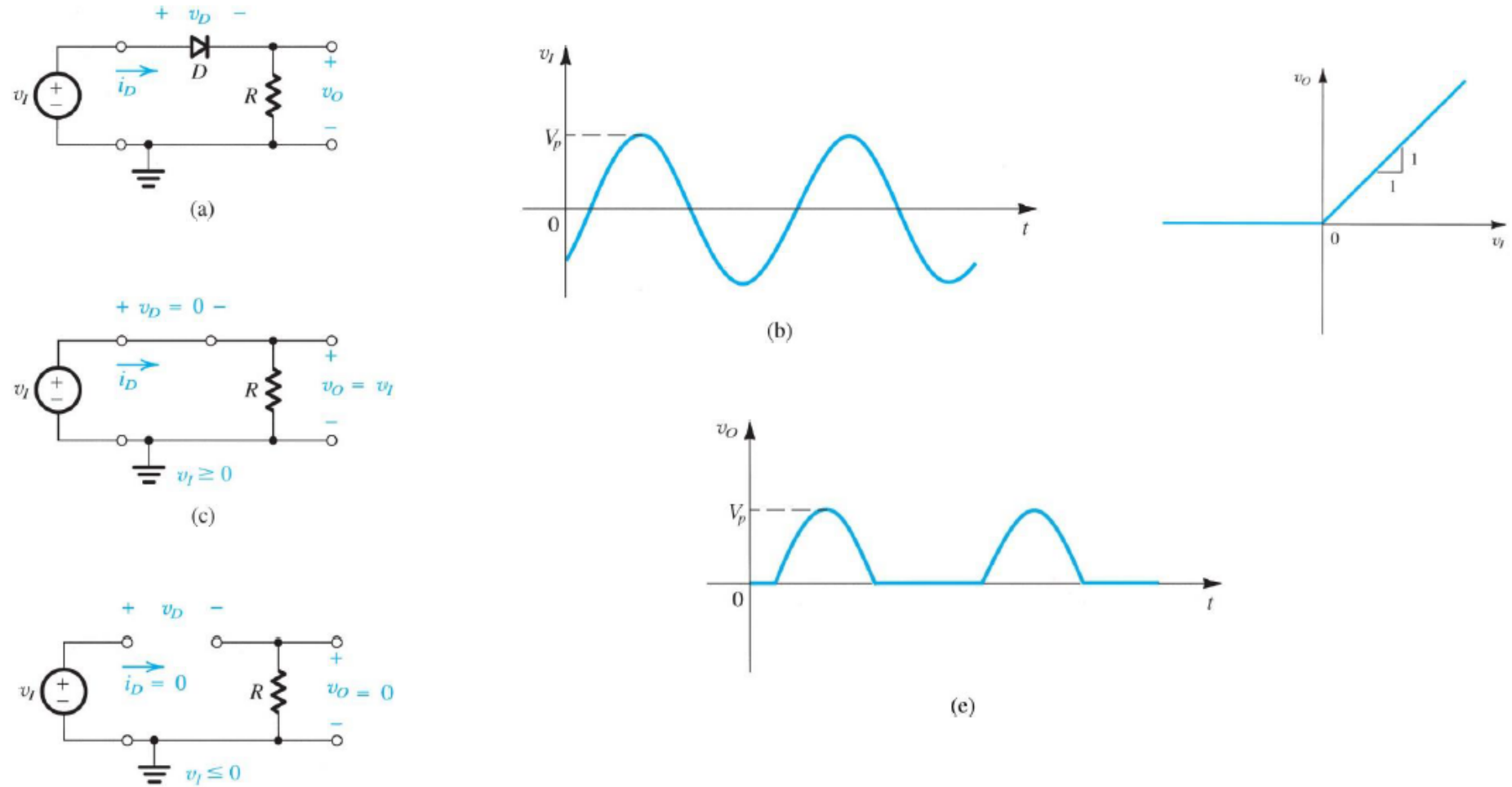
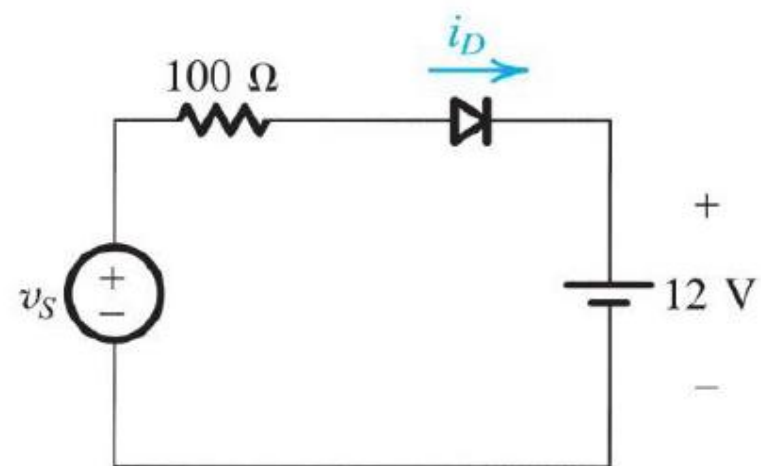
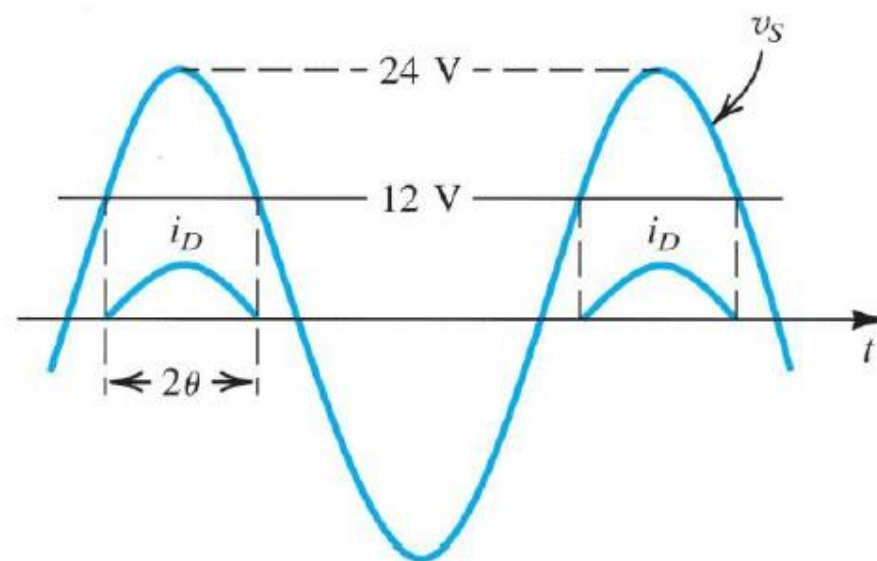


Figure 4.3 (a) Rectifier circuit. (b) Input waveform. (c) Equivalent circuit when $v_I \geq 0$. (d) Equivalent circuit when $v_I \leq 0$. (e) Output waveform.



(a)



(b)

Limiting and Clamping Circuits

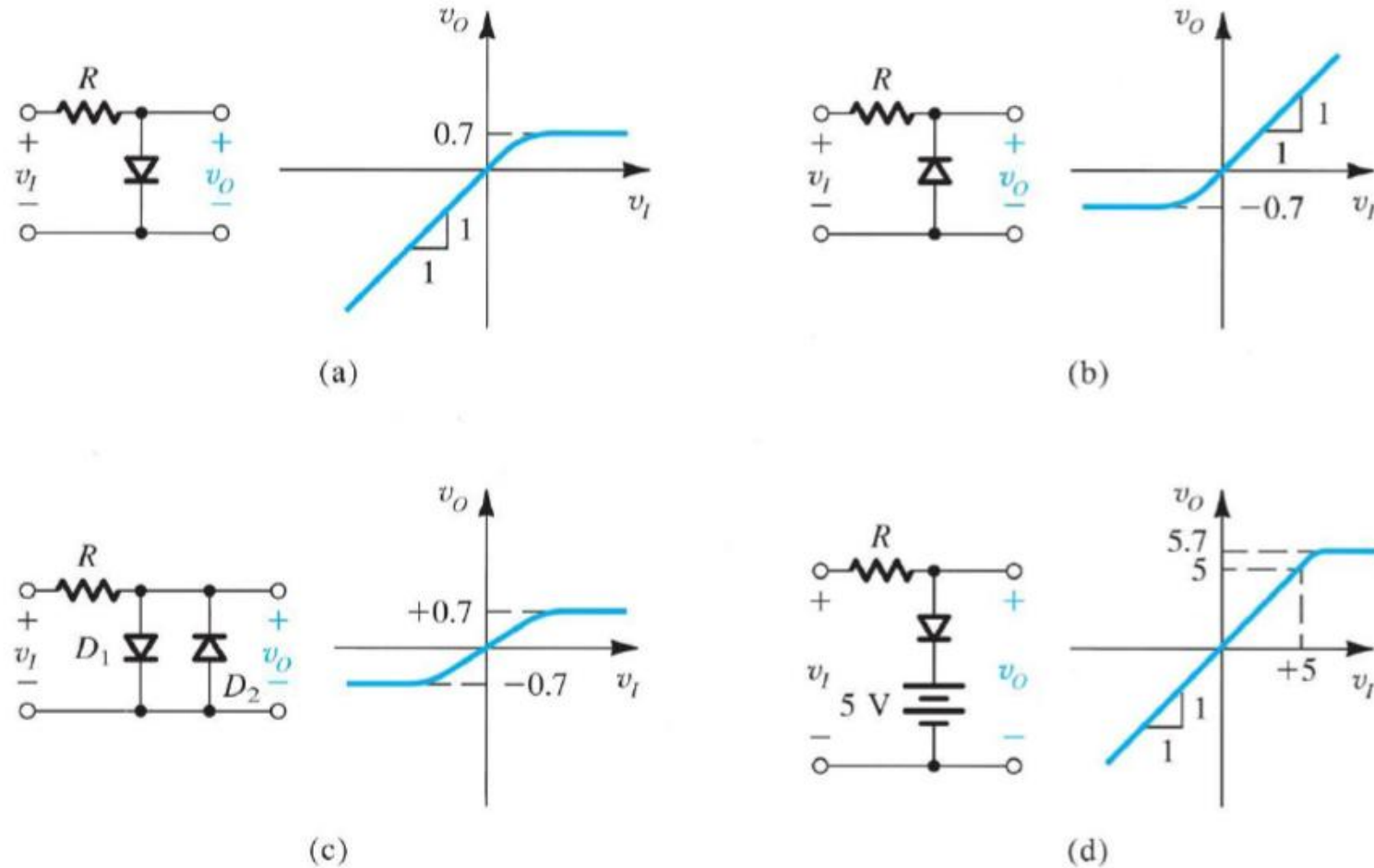


Figure. A variety of basic limiting circuits.

- Assuming the diodes to be ideal, describe the transfer characteristic of the limiter circuit shown in Fig. 4.6 (a).

